Veronica Bolzonella

AI Master's Student
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PROFILE

Driven AI Master's student working at the intersection of deep learning and applied mathematics, in particular dynamical systems, neural differential equations and game theory. Experienced in Neural ODEs and sequence modeling; committed to efficient, sustainable and explainable AI.

EDUCATION

Concordia University, Montreal, Canada — Jan 2027 – June 2027
Master's thesis research, as part of the MSc Artificial Intelligence at Radboud University
- Completing the master's thesis on NeuralGMFG (Neural ODE solvers for Graphon Mean Field Games), funded by the MITACS Globalink Research Award.

Radboud University, Nijmegen, NL — Sept 2025 – 2027 (expected)
MSc Artificial Intelligence (Intelligent Technology), specialization Interactive Agents
- Focus on neural differential equations, dynamical systems, game theory and the foundations of deep learning.
- Coursework: Probabilistic Deep Learning (9.5), Complex Adaptive Systems (9.0), Information Retrieval (8.0), Text and Multimedia Mining (8.5).

Radboud University, Nijmegen, NL — Sept 2022 – July 2025
*BSc Artificial Intelligence, **cum laude** (8.8/10)*
- Coursework: Deep Learning (9.0), Reinforcement Learning (9.5), Multi-Agent Systems (9.0), Linear Algebra (9.5), Calculus I & II (9.5), Bayesian Statistics (8.0), Applied Mathematics (8.5).

Universitas Gadjah Mada, Yogyakarta, Indonesia — Aug 2024 – Dec 2024
Exchange semester, Computer Science: Internet of Things, Smart Environments, Bioinformatics

RESEARCH & PROFESSIONAL EXPERIENCE

Research Intern, Radboud University, Nijmegen — July 2025 – Present
- Investigating Neural Differential Equations for sequence modeling in language; proof of concept for Dynamical Language Models; supervised by Dr. Tal Kachman.

Bachelor Research Thesis, Radboud University — Mar 2025 – June 2025
- *Low Effort Compression for Efficient Diffusion:* analyzed the effects of naive compression on large-scale image generation models (graded 8.5/10).

Student Assistant, Radboud University — 2023 – 2025
- Knowledge Representation (Mar 2025 – June 2025); Formal Reasoning (Nov 2023 – Jan 2024).

Tutor, Recruiter & Project Manager, IB Academy, Utrecht — Oct 2022 – Present
- Delivered 500+ hours of IB Mathematics tutoring.
- Managed recruitment and course materials.
- Co-authored Math study guides.

SKILLS

Programming & tools	Python (advanced); Java, Scala, MATLAB (intermediate); Git, Linux, Jupyter, WandB, LaTeX, Coq
Frameworks	PyTorch, PyTorch Lightning, JAX (Diffrax, Equinox), Hugging Face, TensorFlow, scikit-learn
AI & mathematics	Neural ODEs, deep learning, NLP, multi-agent systems, reinforcement learning, explainable AI, game theory, calculus, linear algebra
Languages	Italian (native), English (fluent), Spanish (working), Portuguese (working), French (basic)

PROJECTS

NeuralGMFG (JAX, Equinox, Diffrax; master's thesis): Neural ODE solvers for Graphon Mean Field Games on multi-community flocking dynamics, validated against analytical Riccati solutions.

Uncertainty Estimation in RAG Systems (Information Retrieval): investigated the accuracy of uncertainty estimation metrics for retrieval-augmented generation, specifically for long-form answers.

Gender and Social Bias in Children's Literature (Text Mining): explored bias in children's books using custom-trained Word2Vec embeddings.

INTERNATIONAL EXPERIENCE & VOLUNTEERING

- **Teacher Assistant, Yogyakarta International School (YIS):** assisted Mathematics and Computer Science courses.
- **AFS Scholarship, Brazil (2019 – 2020):** full immersion year abroad; developed working proficiency in Portuguese.
- **Patmos Foundation, Arusha, Tanzania:** school volunteer and fundraiser manager for local educational initiatives.
- **Erasmus+ Programs:** cross-cultural experience through multiple exchange and training programs.